Model Questions on Discrete Mathematics and Graph Theory (MA203)

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Method of Induction

1- Prove that $n^3 - n$ is divisible by 3 for every positive integer n by using mathematical induction

2- Show that $n^2 > 2n + 1$ for $n \geq 3$ by Mathematical Induction

3- Prove by mathematical induction that $1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}$ for all $n \in \mathbb{N}$

4- Prove by Mathematical induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n.

Warshall's Algorithm

5- Using Warshall algorithm, find the transitive closure of the relation $R = \{(1,2), (2,3), (3,3)\}$ on the set $A = \{1,2,3\}$.

6- Find the transitive closure of the relation $R = \{(1,2), (2,1), (2,3), (3,4)\}$ on the set $A = \{1,2,3,4\}$ using Warshall's algorithm

7- Apply Warshall's algorithm to compute the transitive closure of the relation on A = {a, b, c, d} represented by the matrix $M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

8- Find the transitive closure of the relation $R = \{(2,1), (2,3), (3,1), (3,4), (4,1), (4,3)\}$ on set $A = \{1, 2, 3, 4\}$ using Warshall's algorithm

Solution of Homogeneous Equation by Generating Function, Solution of Non-homogeneous Equation,

9- Solve the recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 0$, for all $n \geq 2$ by using generating function.

10- Solve the recurrence relation $a_{n+2} - a_{n+1} - 2a_n = n^2$

11- Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ for $n \geq 2$, with $a_0 = 1$, $a_1 = 6$ using generating function

12- Find the solution of the recurrence relation $a_n = 2a_{n-1} + n$ for $n \geq 1$, with $a_0 = 1$.

13- Solve $a_n = a_{n-1} + 2a_{n-2}, n \geq 2$ with initial conditions $a_{0=0}, a_1 = 1$.

14-How many solutions are there of $x + y + z = 17$ subject to the constraints $x \geq 1$,$y \geq 2$ and $z \geq 3$

15-How many integers between 1 and 500 (inclusive) are

(i) divisible be at least one of 3, 5, 7?

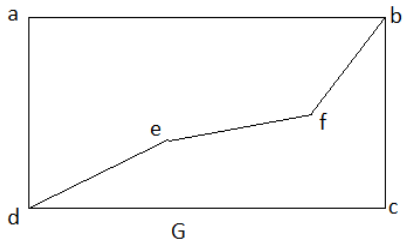
(ii) divisible by 3 and 5 but not by 7?

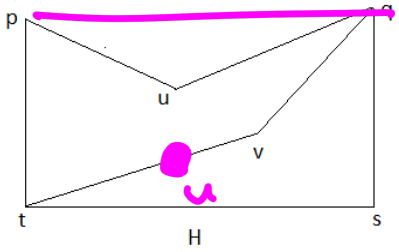
16-A committee of 5 members is to be formed from 6 men and 5 women. In how many ways can this be done when the committee consists of

(i) exactly 2 women, (ii) at least 2 women, (iii) at most 2 women

17-Prove that the number of vertices with odd degree is always even.

18-Test the isomorphism of the following graphs G and H:

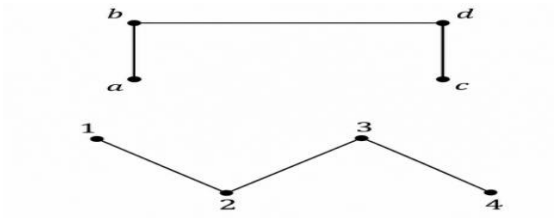




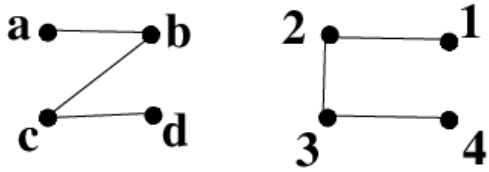
19-Show that Maximum number of edges in a simple graph with n vertices is $\frac{n(n-1)}{2}$

Test the isomorphism of the following two graphs

20-

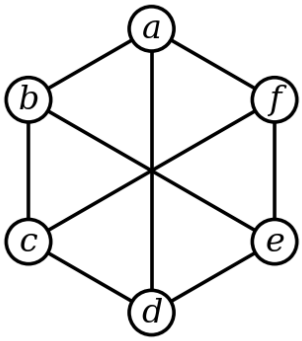


21-Show that the two graphs given below are isomorphic.



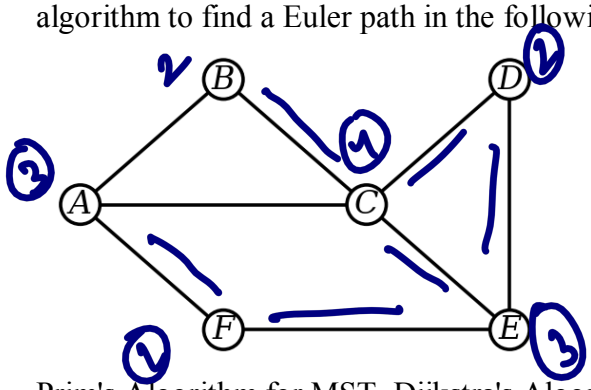
22-State and prove the Handshaking theorem. Hence show that there is no simple graph with degree sequence 3, 3, 2, 2, 1.

23-Define bipartite graph. Check whether the following graph is bipartite or not. If yes, write the bipartition.



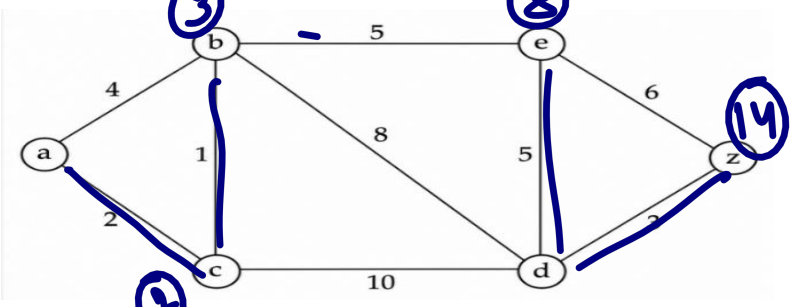
24-State and prove Euler's formula for connected planar graph

25-State the condition for a connected graph to have an Euler path. Use Fleury's algorithm to find a Euler path in the following graph.

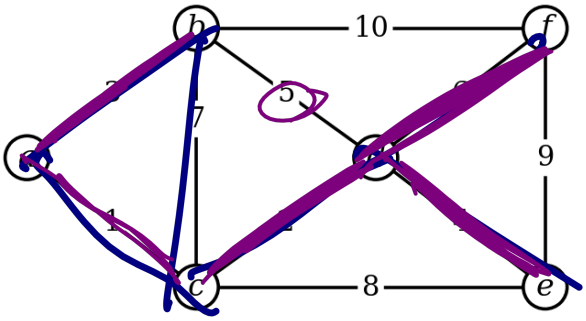


Prim's Algorithm for MST, Dijkstra's Algorithm, Ford-Fulkerson Algorithm.

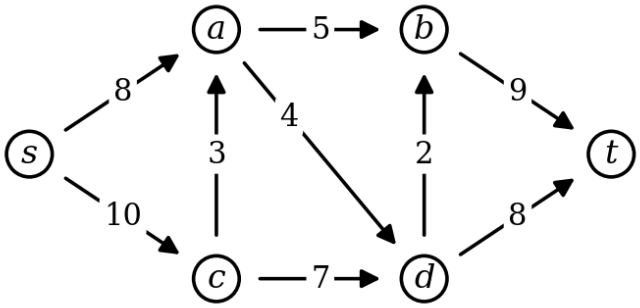
26-Find the length of a shortest path between the vertices a and z by using Dijkstra's algorithm in



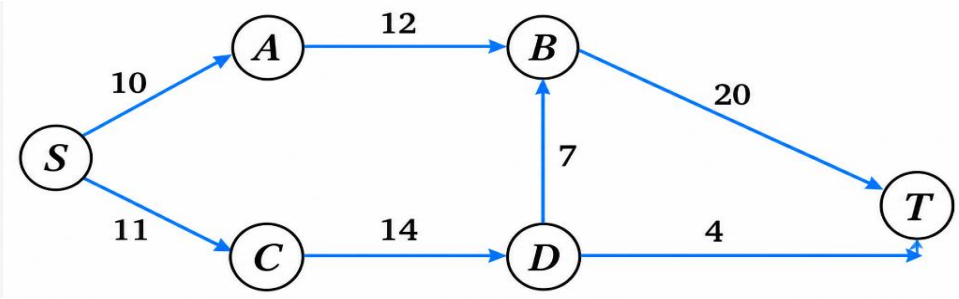
27-Use Prim's algorithm to find a minimal spanning tree (MST) of the following weighted graph, starting from vertex a. Also find its total weight.



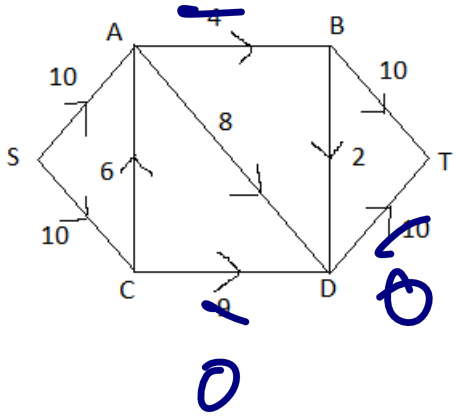
28-Find the maximum flow from s to t in the following network using the Ford-Fulkerson algorithm. Also write the corresponding minimum cut



29-Find the maximum flow of the following network using Ford-Fulkerson algorithm



30-Find the maximum flow of the following network using Ford-Fulkerson algorithm.



Checking whether an Algebraic Structure is a Group and Subgroup, Cosets, Group Codes, Error Detection and Correction.

31-Show that (2 ,5) encoding function defined by e(00)=00000, e(01)=10101, e(11)=11011 form a group code

32-Check whether the algebraic structure $G = (\left\{\begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in Q\right\}; *)$, where $A * B = A.B$ forms a group or not?

33-Let m= 2, n = 5, and

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Determine the group code $e_H: B^2 \rightarrow B^5$.

34-Let $G = (\mathbb{Z}_{12}, +)$ and $H = \{0, 4, 8\}$. Show that H is a subgroup of G. Find all the left cosets of H in G and verify Lagrange's theorem

35-Consider the encoding function $e: B^2 \rightarrow B^5$ defined by $e(00) = 00000, e(01) = 01110, e(10) = 10101, e(11) = 11011$. Show that e is a group code. Find its minimum distance and the number of errors it can detect and correct

36-Let $G = \{(a, b) | a, b \in R, a \neq 0\}$. Define a binary operation $*$ on G by $(a, b) * (c, d) = (ac, bc + d)$ for all $(a, b), (c, d) \in G$, show that $(G, *)$ is a group.

37-Let G be a finite group and H be a subgroup of G, then prove that the order of H divides the order of G

1) $n = 2$

$n^2 - n = 4 - 2 = 2$

which is not divisible by 3

$n = 1$

$1^3 - 1 = 0$ ∴ 0 is divisible by 3

$n^3 - n = n(n^2 - 1) = (n-1)n(n+1)$

$n = 3k$

$n^3 - n = 27k^3 - 3k = 3(9k^3 - k)$

$n = 1$

$n = 3k+1$

$(27k^3 + 27k^2 + 9k + 1) - (3k+1) = 3(9k^3 + 9k^2 + 2k)$

$n = 3k+2$

$(27k^3 + 54k^2 + 36k + 8) - (3k+2) = 3(9k^3 + 18k^2 + 11k + 2)$

2) Put $n = 3$

Left side $= 3^2 = 9$

Right side $2(3) + 1 = 7$

$n = K$

LHS $= K^2$

RHS $\rightarrow 2K + 1$

$n = k+1$

LHS $(k+1)^2 = k^2 + 2k + 1$

RHS $2(k+1) + 1 = 2k + 2 + 1 = 2k + 3$

3) base case $n = 1$

$1 \times 2 = \frac{(1)(2)(3)}{3} = 2$ (H.P)

Case $n = k$

$1 \times 2 + 2 \times 3 + \dots K(K+1) = \frac{K(K+1)(K+2)}{3}$

case $n = k+1$

$1 \times 2 + 2 \times 3 + \dots (k+1)(k+2) = \frac{(K+1)(K+2)(K+3)}{3}$

$\frac{K(K+1)(K+2)}{3} + (K+1)(K+2) = \frac{(K+1)(K+2)(K+3)}{3}$

$(K+1)(K+2)\left(\frac{K}{3} + 1\right)$

$(K+1)(K+2)\left(\frac{K+3}{3}\right)$

LHS = RHS

⑥ $\left\{\frac{6^{n+2}+7^{2n+1}}{43}\right\}$ =

$n=1$

$\frac{6^3+7^3}{43}$ -

$n=k$

$\Rightarrow 6^{k+2}+7^{2k+1} = 43m$

$\Rightarrow 6^{k2} = 43m - 7^{2k+1}$

$n = k\ 1$

$\Rightarrow 6^{1+3} + 7^{2k+3} = 6^{k+2+1} + 7^{2k+1+2}$

$= 6 \cdot 6^{k+2} + 7^2 \cdot 7^{2k+1}$

$= 6 \cdot (43m - 7^{2k+1}) + 43\,(7^{2k+1})$

$= 43 \cdot 6m - 6 \cdot 7^{2k+1} + 49 \cdot 7^{2k+1}$

$= 43\,6m + (49-6)\,7^{2k+1}$

$= 43\,6m + 43 \cdot 7^{2k+1}$

$= 43\,(6m + 7^{2k+1})$ — ②

eqn ② is divisible by 43

(Proved)

7)

	1	2	3	4
1	0	1	0	0
2	1	0	1	0
3	0	0	0	1
4	0	0	0	0

$\{2\} \times \{2\}$

$\{2,2\} \to 1$

0	1	0	0
1	1	1	0
0	0	0	1
0	0	0	0

$\{1,2\} \times \{1,2,3\}$

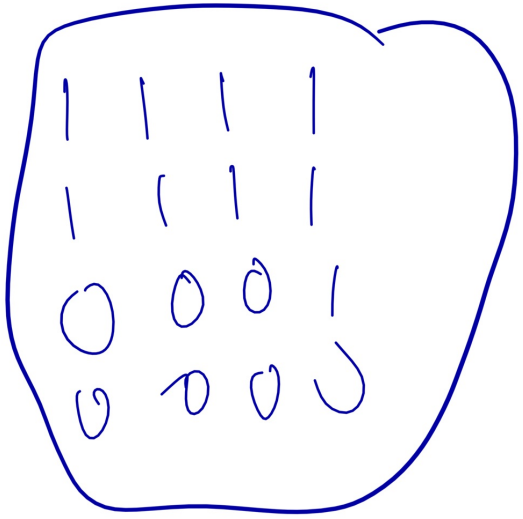
$\{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3)\}$

1	1	1	0
1	1	1	0
0	0	0	1
0	0	0	0

$\{1,2\} \times \{4\}$

$(1,4)\ (2,4)$

1	1	1	1
1	1	1	1
0	0	0	1
0	0	0	0



antwo,

(8)

	1	2	3	4
1	0	0	0	0
2	1	0	1	0
3	1	0	0	1
4	1	0	1	0

Step-1

$\{2,3,4\} \times 0$

$\Rightarrow$

Step-2

$\{0\} \times \{1,3\} \Rightarrow 0$

Step-3

$\{2,4\} \times \{1,4\}$

$(2,1);(2,4);(4,1);(4,4)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Step-4

$\{1,3,4\} \times \{2,3,4\}$

$(1,2);(1,3);(1,4);(3,2);(3,3);(3,4);(4,2);(4,3);(4,4)$

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Ans

5)

	1	2	3
1	0	1	0
2	0	0	1
3	0	0	1

Step 1 = 0

Step-2 : {3} × {2}

(3,2)

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Step-3 : {2,3} × {2,3}

(2,2); (2,3); (3,2); (3,3)

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Ans

	1	2	3	4
1	0	1	0	0
2	1	0	1	0
3	0	0	0	1
4	0	0	0	0

$\{2,3\} \times \{2,3\} = \{(2,2)\}$

0	1	0	0
1	1	1	0
0	0	0	1
0	0	0	0

$\{1,2,3\} \times \{1,2,3\}$

$\{(1,1), (1,2), (1,3)$

$(2,1), (2,2) (2,3)\}$

1	1	1	0
1	1	1	0
0	0	0	1
0	0	0	0

$$9\sum_{n=2}^{\infty} a_n x^n - 6\sum_{n=2}^{\infty} a_{n-1}x^n + 9\sum_{n=2} a_{n-2}x^n = 0$$

$$\left[\sum_{n=0} a_n x^n - a_0 - a_1 x\right] - 6x\left(\sum_{n=2}^{\infty} a_{n-1}x^{n-1} - a_0\right) + 9x^2\left(\sum_{n=2} a_{n-2}x^{n-2}\right) = 0$$

$$G(x) - a_0 - a_1 x - 6x\,G(x) + 6x\,a_0 + 9x^2 G(x) = 0$$

$$G(x)\,[1 - 6x + 9x^2] = a_0 + (a_1 - 6a_0)x$$

$$\frac{x}{(1-x)} = \sum_{n=0}^{\infty} n x^n$$

$$\frac{a_0 + (a_1 - 6a_0)x}{1 - 6x + 9x^2}$$

$9x^2 - 6x + 1 = 0$

$9x^2 - 3x - 3x + 1 = 0$

$3x(3x-1) - 1(3x-1)$

$(3x-1)^2$

$$\frac{a_0 + (a_1 - 6a_0)x}{(1-3x)^2}$$

$$\frac{A}{1-3x} + \frac{B}{(1-3x)^2} = A(1-3x)^2 + B(1-3x)$$

$$A(1 - 6x + 9x^2) + B - 3x = a_0 + (a_1 - 6a_0)x$$

$$A - 6Ax + B - 3Bx = a_0 + (a_1 - 6a_0)x$$

$$A + B - x(6A + 3B) = a_0 + (a_1 - 6a_0)x$$

$a_0 = A + B$

$6A + 3B = a_1 - 6a_0$

$3(A + B) + 3A = a_1 - 6a_0$

$3(a_0) + 3A = a_1 - 6a_0$

$3A = a_1 - 9a_0$

$A = \frac{a_1 - 9a_0}{3}$

$B = a_0 - A$

$a_0 - \frac{a_1 - 9a_0}{3}$

$\frac{3a_0 - a_1 - 9a_0}{3}$

$B = \frac{-a_1 - 6a_0}{3}$

$$\frac{1}{(1-3x)^2} = \sum_{n \geq 0}^{\infty} (n+1)(3x)^n$$

$$\frac{1}{1-3x} = \sum_{n \geq 0}^{\infty} (3x)^n$$

$$A \sum_{n \geq 0}^{\infty} (3x)^n + B \sum_{n \geq 0}^{\infty} (n+1)(3x)^n$$

$$a_n = A(n+1) \cdot 3^n + (a_1 - 6a_0) n \cdot 3^{n-1}$$

10) $a_{n+2} - a_{n+1} - 2a_n = n^2$

This is non homo

$r^2 - r - 2 = 0$

$r^2 - 2r + r - 2 = 0$

$r(r-2) + 1(r-2) = 0$

$(r-2)(r+1) = 0$

$r = 2, -1$

$$a_n^h = C_1 2^n + C_2 (-1)^n$$

$a_n^p = An^2 + Bn + C$

$a_{n+2} = A(n+2)^2 + B(n+2) + C$

$a_{n+1} = A(n+1)^2 + B(n+1) + C$

$a_n = An^2 + Bn + C$

$\Rightarrow A(n^2 + 4n + 4) + B(n+2) + C + A(n^2 + 2n + 1) + B(n+1) + C + An^2 + Bn + C = n^2$

$\Rightarrow$ ~~An^2~~ + ~~$4An$~~ + ~~$4A$~~ + ~~Bn~~ + ~~$2B$~~ + ~~C~~ $- (An^2 +$ ~~$2An$~~ + ~~A~~ $+ Bn +$ ~~B~~ $+ C)$

~~$-(An^2 + Bn + C)$~~

$2An + 3A + B$

$-2An^2 - 2Bn - 2C$

$2An + 3A + B - An^2 - Bn - C$

$$n(2A-2B) - 2An^2 + (3A+B-2C) = n^2$$

$$A = -1/2 \; , \; B = -1/2 \; , \; C = -1$$

$$\boxed{a_n^p = -\frac{1}{2}n^2 - \frac{1}{2}n - 1}$$

$n \geq 2$

$$a_n = 6a_{n-1} - 9a_{n-2} \qquad n \geq 2 \quad a_0 = 1 \; , \; a_1 = 6$$

$$\sum_{n=2}^{\infty} a_n x^n = A(x) - a_0 - a_1 x$$

$$\sum_{n=2}^{\infty} a_{n-1} x^n = x\left(A(x) - a_0\right)$$

$$\sum_{n=2}^{\infty} a_{n-2} x^n = x^2 A(x)$$

$$\sum_{n=2}^{\infty} a_n x^n - 6\sum_{n=2}^{\infty} a_{n-1} x^n + 9\sum_{n=2}^{\infty} a_{n-2} x^n$$

$$A(x) - a_0 - a_1 x - 6x\left(A(x) - a_0\right) + 9x^2 A(x) = 0$$

$$A(x) - a_0 - a_1 x - 6xA(x) - 6xa_0 + 9x^2 A(x)$$

$$A(x)\left(9x^2 - 6x - 1\right) = -a_0 6x + a_0 + a_1 x$$

$$A(x)\left(9x^2 - 6x + 1\right) = -6x + 1 + 6x$$

$$\frac{1}{9x^2 - 6x + 1} = \frac{1}{(3x-1)^2} = \frac{1}{(1-3x)^2}$$

$$\frac{1}{(1-3x)^2} = \sum_{n\geq 0}^{\infty} (n+1)\,3^n x^n = \boxed{a_n = (n+1)3^n}$$

$$\boxed{\frac{1}{(1-rx)^2} = \sum_{n=0}^{\infty} (n+1)\,r^n x^n}$$

12) $2a_{n-1} + n \quad n \geq 1, a_0 = 1$

$a_n - 2a_{n-1} = n$

$x - 2 = 0$

$x = 2$

$a_n^c = C \cdot 2^n$

$An + B - 2(A(n-1) + B) = n$

$An + B - 2(An - A + B) = n$

$An + B - 2An + 2A - 2B = n$

$n(A - 2A) + (B + 2A - 2B) = n$

$A = -1 \qquad 2A - B = 0$

$-2 - B = 0$

$B = -2$

13) $a_n = a_{n-1} + 2a_{n-2} \quad n \geq 2, a_0 = 0, a_1 = 1$

$a_n - a_{n-1} - 2a_{n-2} = 0$

$\sum_{n=2}^{\infty} a_n x^n - \sum_{n=2}^{\infty} a_{n-1} x^n - 2\sum_{n=2}^{\infty} a_{n-2} x^n = 0$

$(G(x) - a_0 - a_1 x) - x(G(x) - a_0) - 2x^2 G(x) = 0$

$(G(x) - 0 - x) - x(G(x)) - 2x^2 G(x) = 0$

$G(x) - x - xG(x) - 2x^2 G(x) = 0$

$G(x)[1 - x - 2x^2] = x$

$G(x) = \frac{x}{(x+1)(2x-1)} = \frac{x}{(x+1)(2x-1)}$

$\frac{A}{x+1} + \frac{B}{2x-1}$

$2Ax - A + Bx + B = x$

$x(2A + B) + (B - A) = x$

$B - A = 0$

$B = A \qquad B = 1/3$

$2A + B = 1 \rightarrow 3A = 1 \quad A = 1/3$

$\frac{1}{3}\ \frac{1}{1+x} + \frac{-1}{3}\ \frac{1}{1-2x}$

$$G(x) = \frac{1}{3}\sum_{n=0}^{\infty}(-1)^r x^r - \frac{1}{3}\sum_{n\geq 0}(2x)^r$$

$$a_r = \frac{1}{3}(-1)^r - \frac{1}{3}2^r$$

12) $x + y + z = 17$

$x \geq 1$ } $x - 1 \geq 0 = a$

$y \geq 2$ } $y - 2 \geq 0 = b$

$z \geq 3$ $z - 3 \geq y = c$

$$(x' + 1) + (y' + 2) + (z' + 3) = 17$$

$$x' + y' + z' = 17 - 16 = 11$$

for $x_1 + x_2 + x_3 + \cdots + x_r = r$

$$\binom{n + r - 1}{r - 1} \quad (\text{Here } n = 11\ r = 3)$$

$$\binom{11 + 3 - 1}{3 - 1} = \binom{13}{2}$$

$$\frac{13 \times 12}{2}, \to 78$$

15) 500/3 = 166
500/5 = 100
500/7 = 71

166+100+71 = 337

divisible 3 & 5 = 15 → 500/15 = 33
divisible 3 & 7 = 21 → 500/21 = 23
divisible 5 & 7 = 35 → 500/35 = 14

337 − 33 − 23 − 14 = 267

divisible by 3, 5, 7 = 105
500/105 = 4

267 + 4 = 271

(ii) $\frac{500}{15} = 33$ 15 = LCM of 3 & 5

105 (LCM of 3 & 5 & 7)

$\frac{500}{105} = 4$

33 − 4 = 29

16) $^5C_2 \times \frac{5!}{2!\,3!} = 10$ $\quad$ $^6C_3 \times \frac{6!}{3!\,3!} = 20$

$10 \times 20 = 200$

Q) 2 women

→ 2 women & 3 men
$^5C_2 \times {}^6C_3 = 10 \times 20 = 200$

→ 3 women & 2 men
$^5C_3 \times {}^6C_2 = 10 \times 15 = 150$

→ 4 women & 1 men
$^5C_4 \times {}^6C_1 = 5 \times 6 = 30$

→ 5 women
$^5C_5 \times {}^6C_0 = 1 \times 1 = 1$

200 + 150 + 30 + 1 = 381

No women 5 men
$^6C_5 = 1 \times 6 = 6$

1 women, 4 men
$^5C_1 \times {}^6C_4$
$5 \times 15 = 75$

2 women, 3 men
$^5C_2 \times {}^6C_3 = 10 \times 20 = 200$

$200 + 75 + 6 = 281$

(17) $\sum_{v \in V} \deg(v) = 2|E|$

$3 + 5 = 8$

$3 + 5 + 7 = 15$

Sum of all odd degree vertices is even

(18)

a	2
b	3
c	2
d	3
e	2
f	2

p	2
q	3
s	2
t	3
u	2
v	2

$a \to p$
$b \to q$
$c \to s$
$d \to t$
$e \to u$
$f \to v$

non-isomorphism

(19) $\frac{1}{2} = \frac{n!}{2!(n-2)!} \Rightarrow \frac{n(n-1)(n-2)!}{2(n-2)!} \Rightarrow \frac{n(n-1)}{2}$

22) $\sum_{v \in V} \deg(v) = 2|E|$

$3+3+2+2+1 = 11$

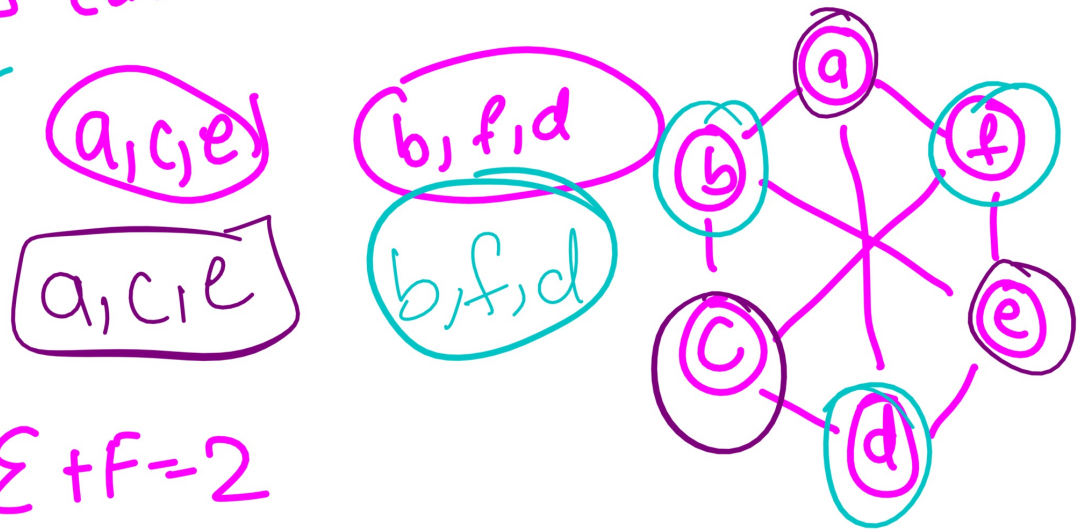
$11 = 2|E|$

$|E| = \frac{11}{2} = 5.5$

No simple graph

23) Vertices can be divide can 2 disjoint

One edge should not connected to another set of edge



24) $V - E + F = 2$

V = No of vertex

E = No of edges

F = No of faces

Graph remain connected

Edge decrease 1

Vertices remain change

25) Zero odd degree vertex ice circuit
two odd degree vertex path

Euler path exist

A → B → C → A → F → E → D → C → E

Complete connection node

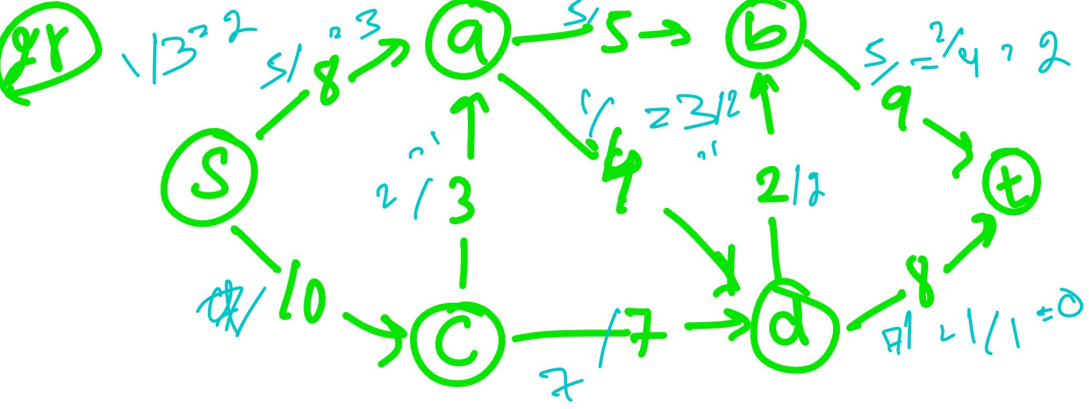
26) a → c → b → e → z

2 + 1 + 5 + 6 = 14

27)

a - c	1	a, c
c - d	2	a, c, d
a - b	3	a, b, c, d
d → e	4	a, b, c, d, e
d → f	6	a, b, c, d, e, f

28)



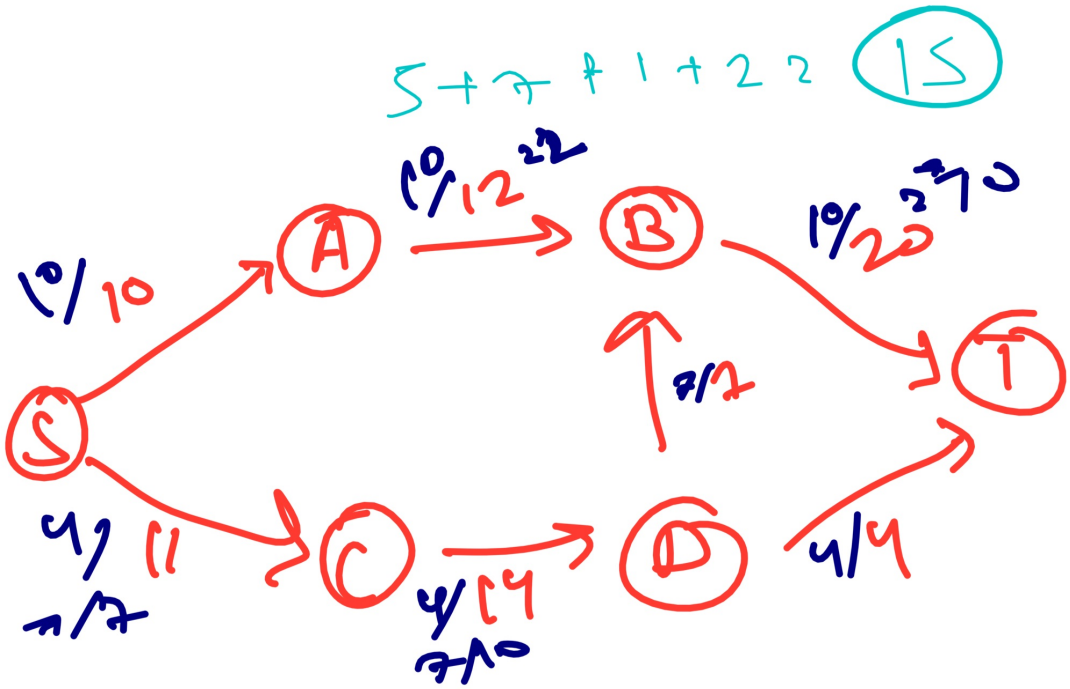
$S \to a \to b \to t = 5$

$S \to c \to d \to t = 7$

$S \to a \to d \to t = 1$

$S \to c \to a \to d \to b \to t = 2$

$5 + 7 + 1 + 2 = 15$

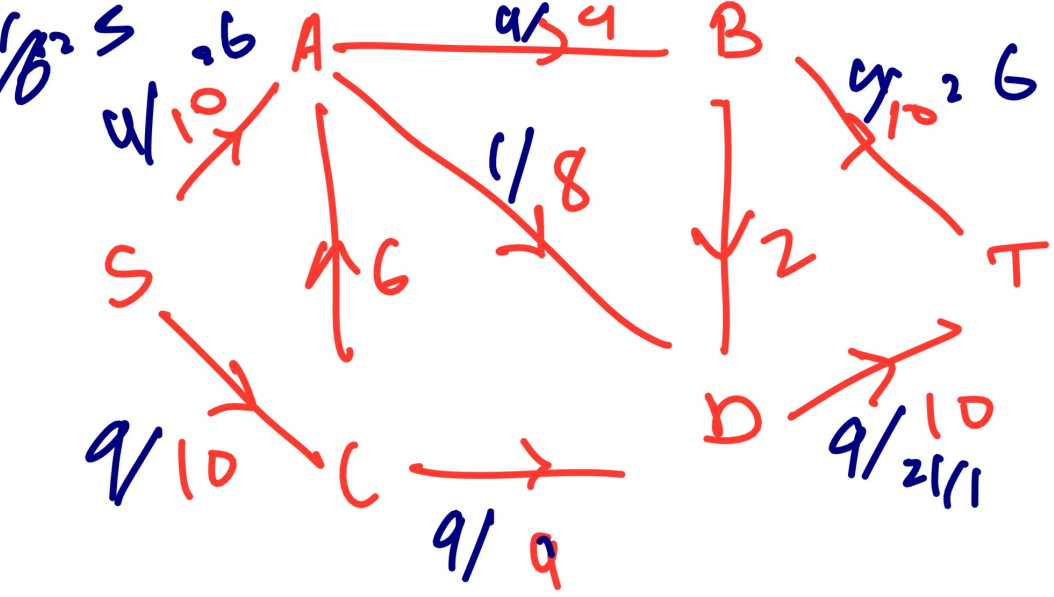


$S \to C \to t = 4$

$S \to A \to B \to T = 10$

$S \to C \to D \to B \to T = 7$

Ответ: 21



$S \to A \to B \to T = 4$

$S \to C \to D \to T = 9$

$S \to A \to D \to T = 1$

(4) N

(3) $e(0,0) = 200000$

$e(0,1) = 10101$

$e(1,1) = 11011$

$e(0,0) \oplus e(0,1) = 10101$

$e(0,1) \oplus e(1,1) = 01110$

$e(0,0) \oplus e(1,1) = 11011$

$\oplus = 01110$

A =

32) closer ABEG then A·BEG A·B = C & all element are hence proved

Associativity = A,B,C ∈ G $(A*B)*C = A*(B*C)$

Identify $A*e = A$

$e = A \cdot A^{-1} = I$

Inverse $A \cdot A^{-1} = I$

Nonzero matrix

33) $B^2 = \{(00),(01),(10),(11)\}$

B^{S_2} ____ - - - -

$e_H(00) = 00\ x_1\ x_2\ x_3$

$x_1 = b_1 h_{11} + b_2 h_{21} = 0$

$x_2 = b_1 h_{12} + b_2 h_{22} = 0$

$x_3\ b_1 h_{13} + b_2 h_{23} = 0$

$e_H(00) = 00000$

$e_H(01) = 0.1\ x_1\ x_2\ x_3$

$x_1 = b_1 h_{12} + b_2 h_{21} = 0 + 1 \cdot 1 = 1$

$x_2 = b_1 h_{12} + b_2 h_{22} = 0 + 1 \cdot 1 = 1$

$x_3 = b_1 h_{13} + b_2 h_{23} = 0 + 1 \cdot 1 = 1$

0 1 1 1

$C_H(1,0) = 1\ 0 \quad n_1\ n_2\ n_3$

$n_1 = b_1 h_{11} + b_2 h_{21} = 1 + 0 = 1$ **1 0 1 1 0**

$n_2 = b_1 h_{12} + b_2 h_{22} = 1 + 0 = 1$

$n_3 = b_1 h_{13} + b_2 h_{23} = 0 + 0 = 0$

$C_H(1,1) = 1\ 1 \quad n_1\ n_2\ n_3$

$n_1 = 1$

$n_2 = 1$ **1 1 1 1 1**

$n_3 = 1$

34) Closure $a, b \in H$ $\quad a+b \in H$

$4+4 = 8 \in H$

$4+8 = 12\ \%\ 12 = 0 \in H$

$8+8 = 16\ \%\ 12 = 4 \in H$

Identity = Identity of G is 0 $\quad 0 \in H$

Inverse: $a * a^{-1} = 0$

$\text{Inv}(0)$ is $0 \in H$

$inv(4) = 12 - 4 = 8 \in H$

$inv(8) = 12 - 8 = 4 \in H$

Hence H is a subgrp of G

$0 + H = \{0, 4, 8\}$

$1 + H = \{1, 5, 9\}$

$2 + H = \{2, 6, 10\}$

$3 + H = \{3, 7, 11\}$

Lagrange theorem

$|G| = |H||G:H| \Rightarrow 12 = 3 \times 4$

35) $e(00) = 00000$

$e(01) = 01110$

$e(10) = 10101$

$e(11) = 11011$

mutative $a*b \in G$, $a*b \in G$

	00000	01110	10101	11011
00000	00000	01110	10101	11011
01110	01110	00000	11011	10101
10101	10101	11011	00000	01110
11011	11011	10101	01110	00000

Associativity: $(a*b)*c = a*(b*c)$

identity: $a*e = a$

$01110 \oplus 00000 = 01110$

$00000 \in G$ hence identity

Inverse $a*a^{-1} = e$

$x \oplus x^{-1} = 00000$

$x = x^{-1}$

distance $00000 \oplus 10101 = 3$ min

$00000 \oplus 01102 = 4$

(36) commutative $(a,b)*(c,d) = (ac, bc+d)$

$ac \neq 0,\ bc+d \in \mathbb{R}$

Hence

Associativity :- $(a,b), (c,d), (e,f) \in G$

$((a,b)*(c,d))*(e,f) = (a,b)*((c,d)*(e,f))$

$(a,b)*(ce, de+f) = (ace, bce+de+f)$

Identity $= a*c = a$

$(a,b) * I = (a,b)$

Let $I = (x,y)$

$(a,b) * (x,y) = (ax, bx+y)$

$ax = a,\ x \neq 1$ $\quad bx+y = b$

$y = 0$

Inverse - $a * a^{-1} = e$

$(a,b) * (g,h) = (1,0)$

$(ag, bg+h) = (1,0)$

$ag = 1$ | $bg+h$

$g = \frac{1}{a}$ | $h = -bg$

$h = -b/a$

$(g,h) \in G \quad a \neq 0$

$(ab)^{-1} = \left(\frac{1}{a}, \frac{-b}{a}\right)$

(37) $|H| \nmid |G|$

$a \in G$

$aH = \{ah : h \in H\}$

$|G| = \underbrace{|H| + |H| \cdots\cdots |H|}_{k \text{ times}}$

$|G| = k|H|$

$\frac{|G|}{|H|} = k$

60-65